



Using Open Mapping to Tackle Flood Risks and Build a Climate-Ready Nakuru City, Kenya

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*Flood Risk Assessment
Maps showing flood-prone
areas in Nakuru City.*

SUMMARY

In the city of Nakuru, Kenya, rapid urban growth and disrupted natural water flows have made flooding a persistent threat. Through the Imagery for Social Good initiative—supported by HOT’s Eastern and Southern Africa Hub and Microsoft—the Nakuru city government and local communities came together to map flood-prone areas and understand the risks. With newly available open geospatial data and training, the city has begun designing nature-based solutions like sponge infrastructure, reducing the threat of displacement for thousands of residents.

UNDERSTANDING THE CONTEXT

Kenya is increasingly vulnerable to climate change, with frequent and severe flooding, droughts, and temperature extremes disrupting livelihoods and urban infrastructure. World Bank reports that over 70% natural disasters in Kenya are linked to extreme climatic events, with flooding alone affecting an estimated 150,000 people each year (World Bank Group, [2019](#), [2021](#)).

The city of Nakuru, one of the country’s fastest-growing urban centers, is particularly at risk.

Situated in the Great Rift Valley, between the Mau and Aberdare mountain ranges, Nakuru receives heavy stormwater during rainy seasons. Moreover, the population has grown tremendously over the years, driving housing demand up. This rapid urbanization has increased stormwater runoff, leading to flooding in the lower parts of Nakuru.

Mr. Festus Bor, City Planner from the Nakuru City Government, explains that due to flooding, several communities near Lake Nakuru had to relocate, leaving their homes and their livelihoods behind.

¹ All research, analysis, and conclusions are the authors’ own. OpenAI was used to assist with drafting, and the text was reviewed, edited and validated by humans.

The Nakuru Government recognized these challenges and has been working on addressing them even before 2021, when Nakuru earned city status. For example, it has been preparing a County Spatial Plan, which aims to improve urban and rural development ([Nakuru County Government, 2019](#)). However, Mr. Bor acknowledged that the County, including the city of Nakuru, had limited baseline geodata that hindered effective flood risk mitigation planning.

OUR APPROACH IN ACTION

In 2022, HOT's Eastern and Southern Africa Open Mapping Hub, Microsoft, and the Nakuru City Government collaborated to harness Machine Learning (ML) technologies to generate open and accessible geospatial data.

Anchoring back to the concept of “sponge city”, where urban features allow water to be absorbed and filtered below-ground ([Sieker, n.d.](#); [World Future Council, 2016](#)), the Satellite Imagery for Social Good Kenya project aimed to use the data to support the city's flood mitigation strategies and strengthen its overall climate resilience.

“The team got training from HOT on how to map on OSM, extract data from QGIS, and use other mapping tools that made the team more efficient in the use of geospatial data”

Mr. Gitau Thabanja
Nakuru's City Manager

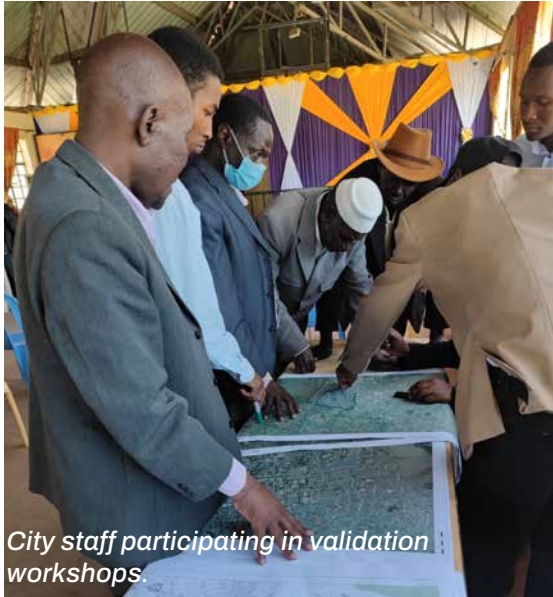
Field data collection activity.



Activities included training, AI-assisted remote mapping, field mapping, and flood risk modelling and verification. The training involved twenty city staff and OpenStreetMap (OSM) community members learning to use AI-enabled tools and data, such as [MapwithAI](#), [Rapid](#), and [Microsoft AI's open buildings dataset](#) (Microsoft, 2022)

It also included an introduction to OSM's remote mapping tools, in addition to field data collection tools, like ODK, OSMAnd, and StreetComplete, necessary to add extra attributes such as street names, schools, and other points of interest to OSM.

AI-assisted mapping enabled quicker remote mapping and provided more time for human validation and participatory field mapping. Local communities were then engaged to help identify high-risk zones, verifying flood models with lived experience.



City staff participating in validation workshops.

The terrain model HAND (Height Above Nearest Drainage) was then employed to simulate flood scenarios and prioritize interventions.

The collaboration generated **1.4 million building footprints, 19,131km of roads, and over 4,000 village names being added to OSM, covering nearly 21,000 km². Data was then used to develop a flood risk assessment of Nakuru City** to better inform the city's flood mitigation strategy (HOT's Eastern and Southern Africa Open Mapping Hub, 2022).

IMPACT ON PEOPLE AND COMMUNITIES

The data revealed flood-prone areas and surfaced gaps in the drainage infrastructure. **Mr. Gitau Thabanja explained how the generated baseline data and flood risk assessments helped the city prioritize investments for upgrading its drainage systems and develop a technical proposal for a sponge city model**—featuring nature-based solutions like permeable pavements and green spaces.

With improved flood modeling, Nakuru began closing unsafe open drainage channels and addressing solid waste accumulation that blocked water flow. Insights from the data allowed planners to target high-risk areas and allocate resources more efficiently.

Mr. Bor adds that **about 10,000 people indirectly benefited from the improved drainage systems**. In the past, other families chose to relocate from their homes, with floodwaters damaging belongings, disrupting work, and school.

Now, with the construction of new stormwater drainage infrastructure in the flood-prone areas, the city has seen people return to the communities. Residents are better able to focus on rebuilding their lives and livelihoods.

Participatory mapping also fostered stronger collaboration between city officials and residents.

“For almost 100 million Kenya[n] shillings, we’ve done two main drainages. And I’m happy to report that this time around, even with heavy rains, we have not experienced [the same level of] flooding as before. Through this mapping, we have now started addressing the challenges of flooding not from guesswork, but from informed GIS-based guidance.”

Mr. Gitau Thabanja,
Nakuru's City Manager

Nakuru's efforts have been recognized by the World Bank through its City Planning Labs (CPL) program (Nakuru County Government, 2024; World Bank Group, n.d.), validating the city's commitment to data-driven and collaborative urban planning. The award also helped advocate for increased funding for GIS and resilience infrastructure.

HOW CHANGE CAN LAST

Nakuru City continues to integrate open geospatial data into its urban planning strategies. Building on the success of the Satellite Imagery for Social Good project, the city has joined HOT's Climate Ready Cities Program (HOT's Eastern and Southern Africa Open Mapping Hub, 2024).

Beyond mitigating flood risks, the city is now using GIS to inform its sustainable urban mobility efforts (Nakuru County Government, 2024). This includes planning safer infrastructure through walkways and cycling lanes, supporting green infrastructure. They further strengthened their Nakuru Sponge City Program using data to inform nature-based solutions to address urban and climate challenges (Holmes, 2025; Kenya News Agency, 2024).

"Through [a] participatory approach, we are able to create a pool of residents that we can engage in not only in the flooding but also for public participation on budget making. This process gives us a community that we can rely on."

Mr. Gitau Thabanja,
Nakuru's City Manager

Notably, the participatory approach has also improved how the city systematically involves its residents in shaping policies and projects. Community members now contribute to identifying local risks, leading to stronger civic engagement and more responsive planning.

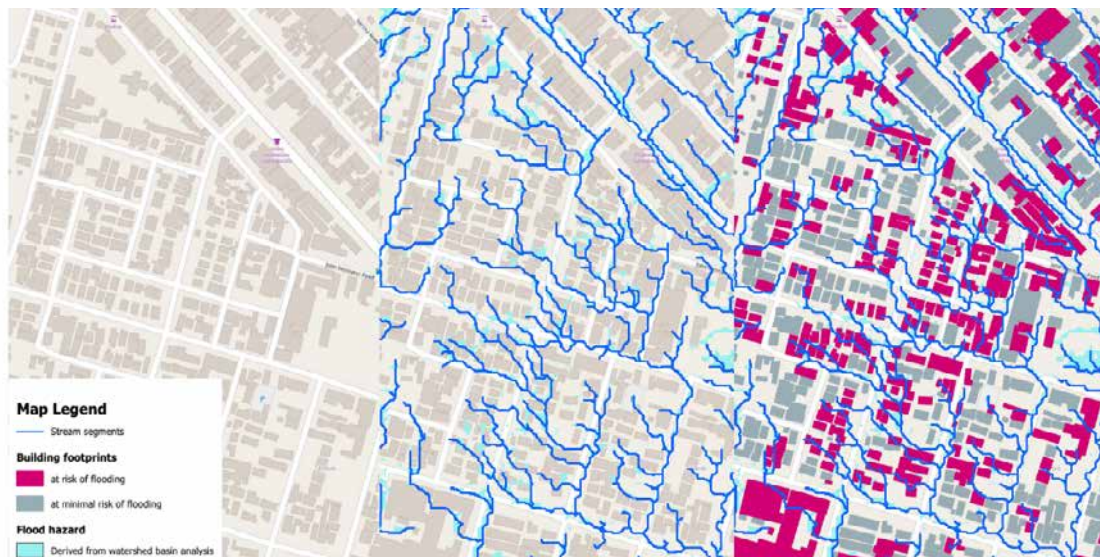
By embedding geospatial tools, data, and participatory processes into daily operations, Nakuru is laying the groundwork for more inclusive and sustainable urban governance.



*The Nakuru City team announcing they received the World Bank City Planning Labs (CPL) grant.
Source: Nakuru County Government, 2024.*

INSIGHTS FOR THE FUTURE

This is an exemplary case of how combining AI-assisted tools with local mapping and community collaboration can drive data-informed, sustainable urban development. By integrating these efforts with institutional commitment, the city is both responding to current climate challenges and laying a strategic foundation for long-term urban resilience. Nakuru shows what can happen when decision-makers understand the possibilities brought by open geospatial data and tools, and communities are deliberately engaged with for urban improvement actions.



Resulting Flood Risk Assessment Map

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